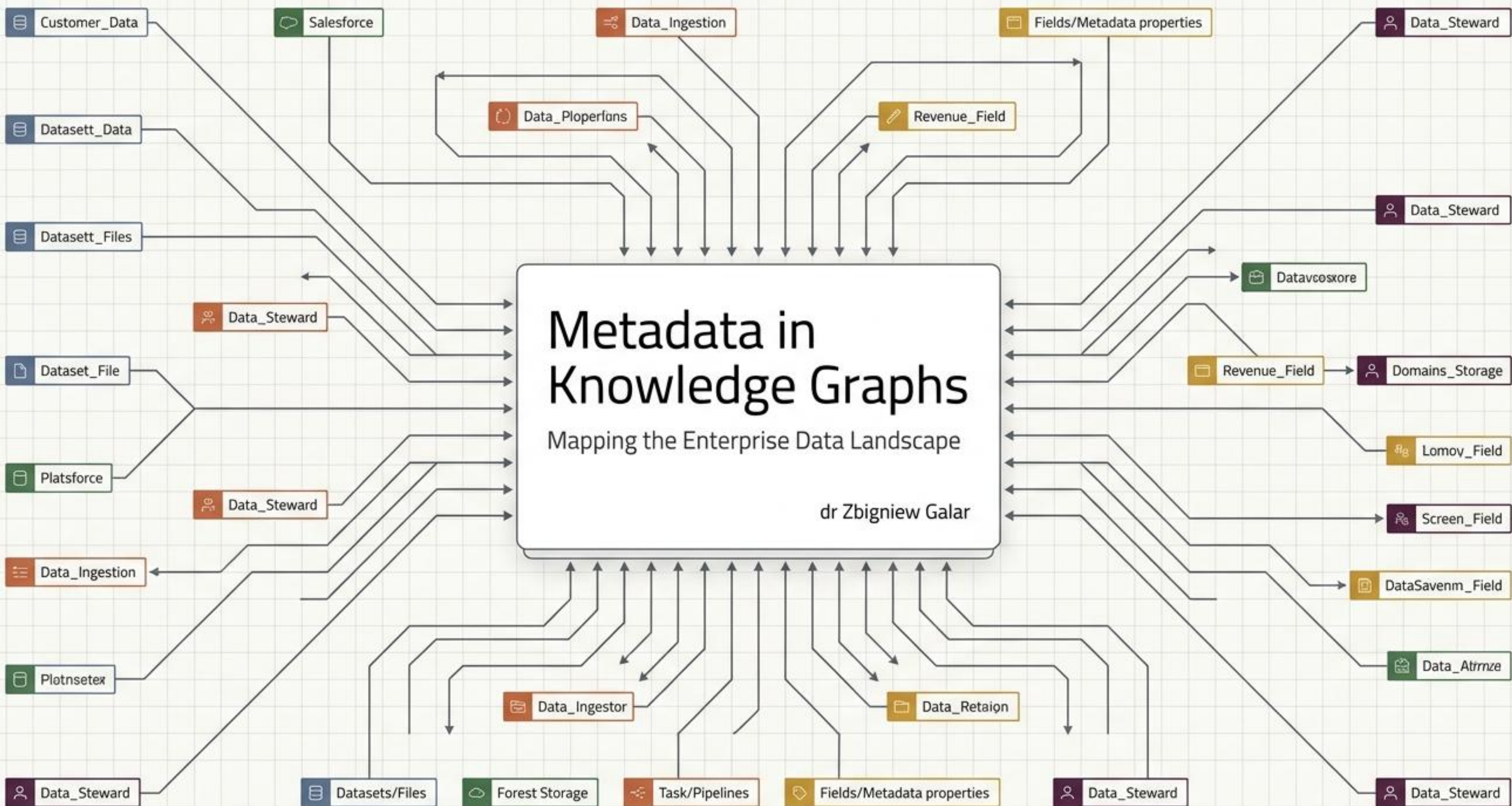
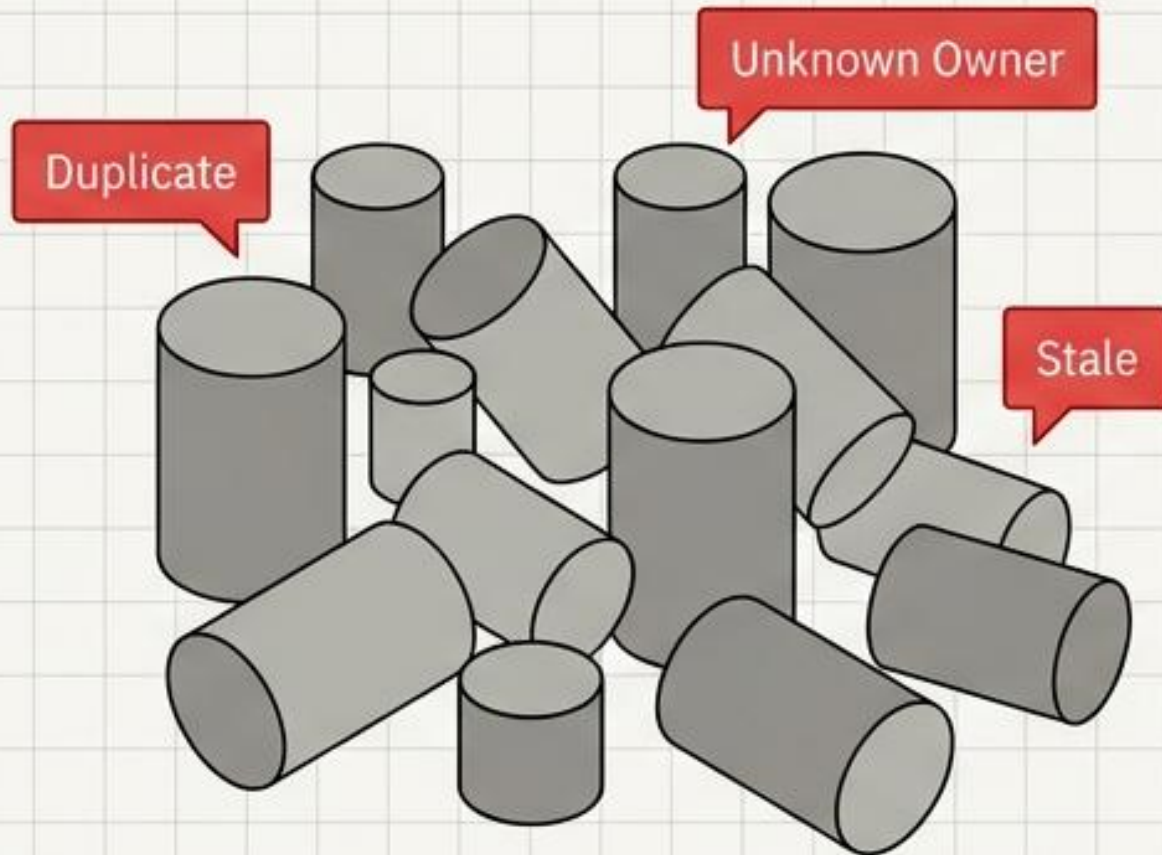




THE PROBLEM



Fragmented Silos:

Identical data duplicated across isolated departmental platforms.

Broken Provenance:

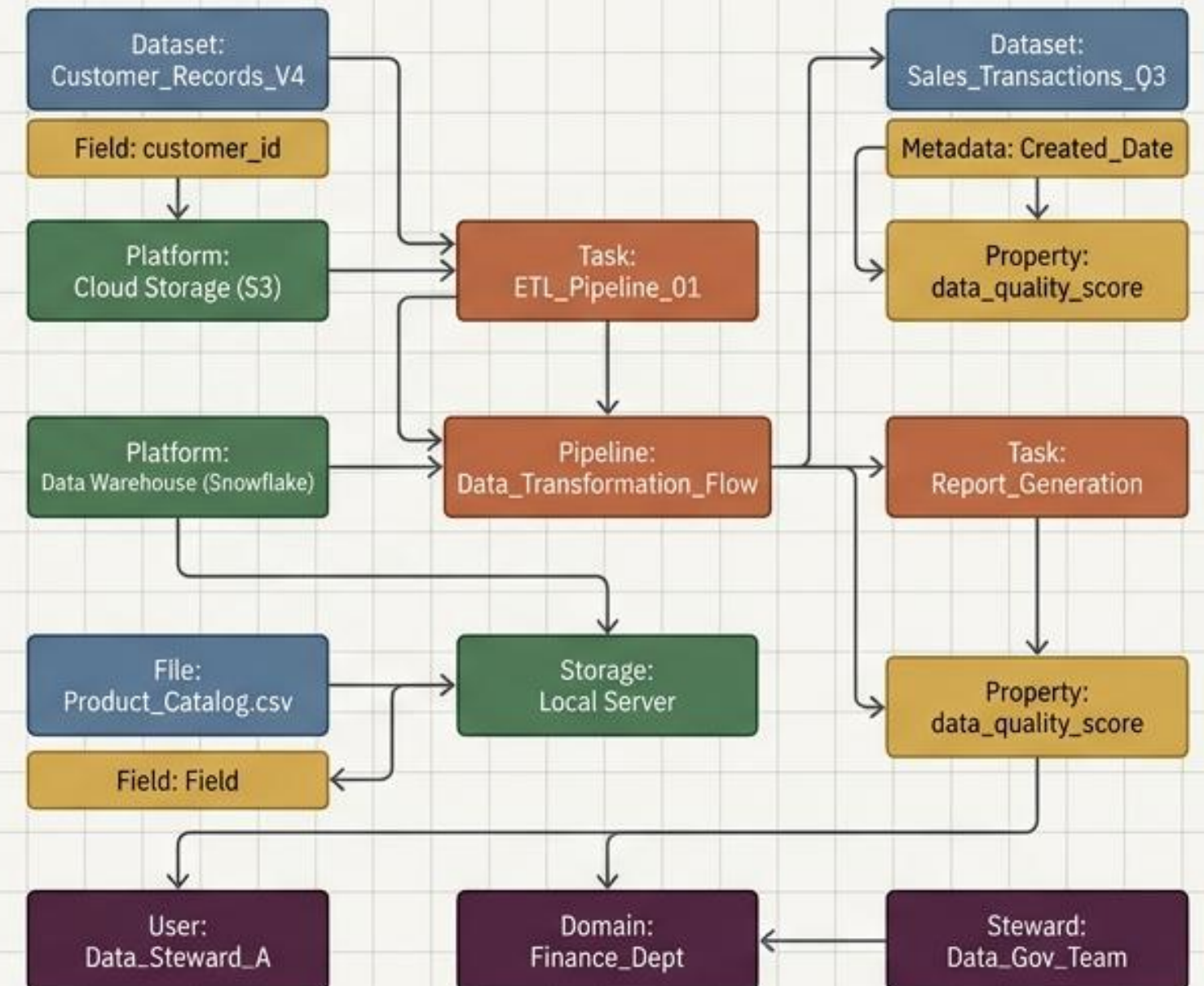
Total inability to trace a data asset's origin or transformation history.

Blind Consumption:

Zero visibility into the downstream impact when source data changes.

THE IDEAL

The Unified Metadata Graph

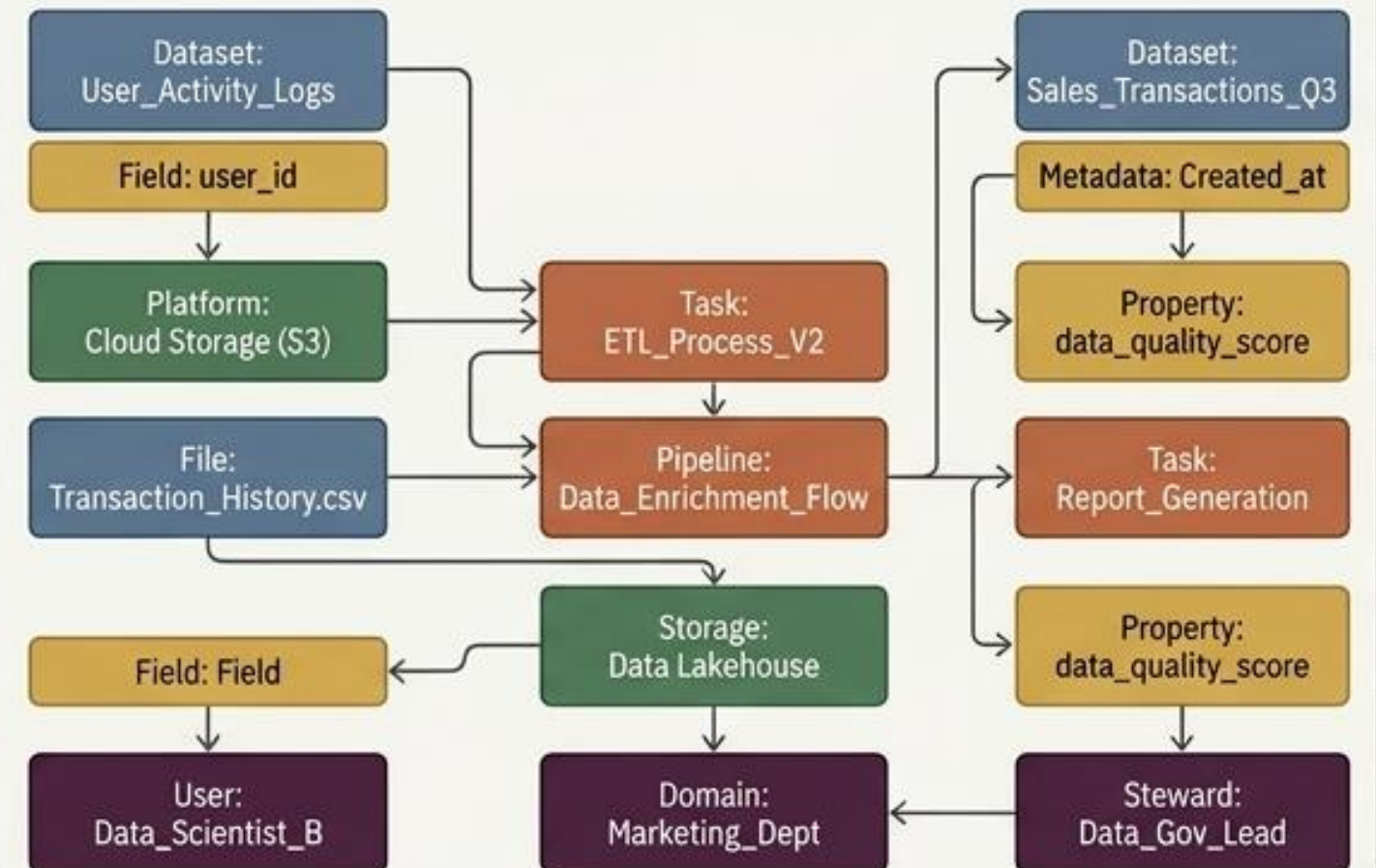


The Rise of the Metadata Hub

2017: The Inflection Point



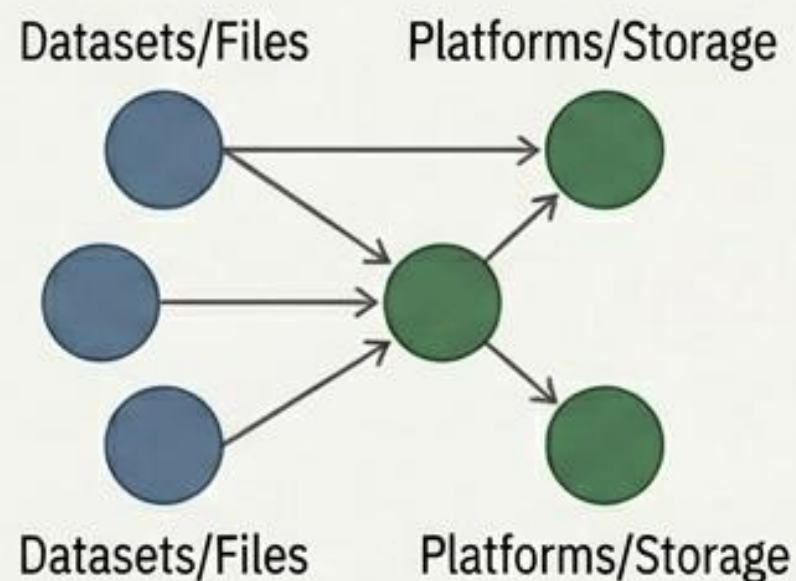
The Graph-Backed Metadata Hub



The Critical Transition: Moving from passive, static data dictionaries to active, graph-based data discovery engines.

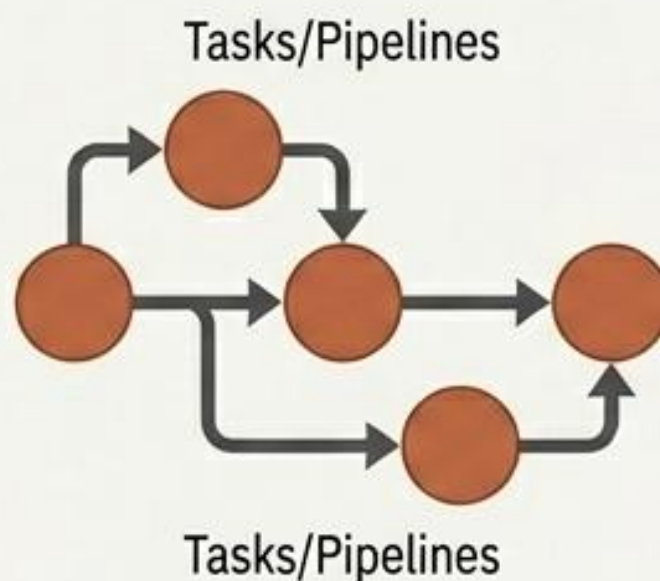
Defining the Metadata Knowledge Graph (MKG)

Shape & Location



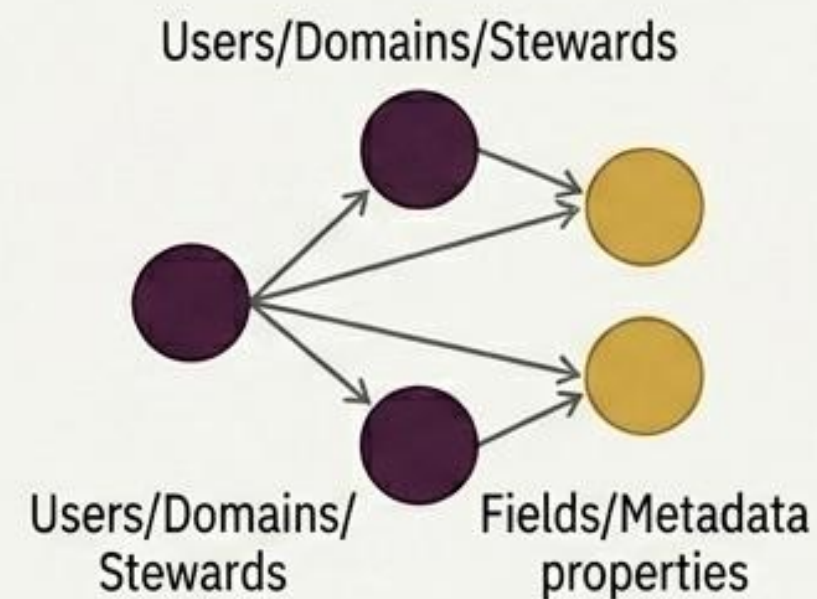
Where data lives and its physical/logical schema.

Process & Provenance



The actions, pipelines, and tasks that transform the data.

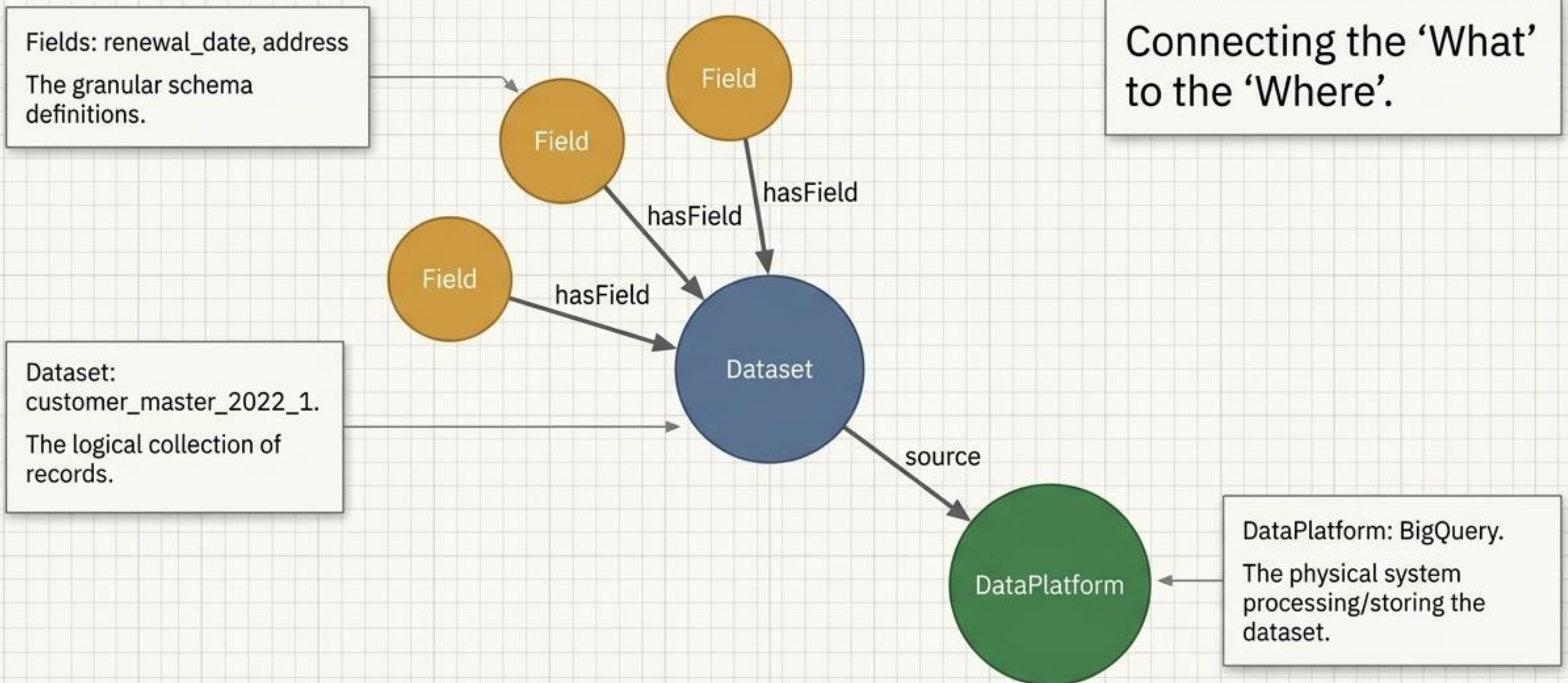
Consumers & Sinks



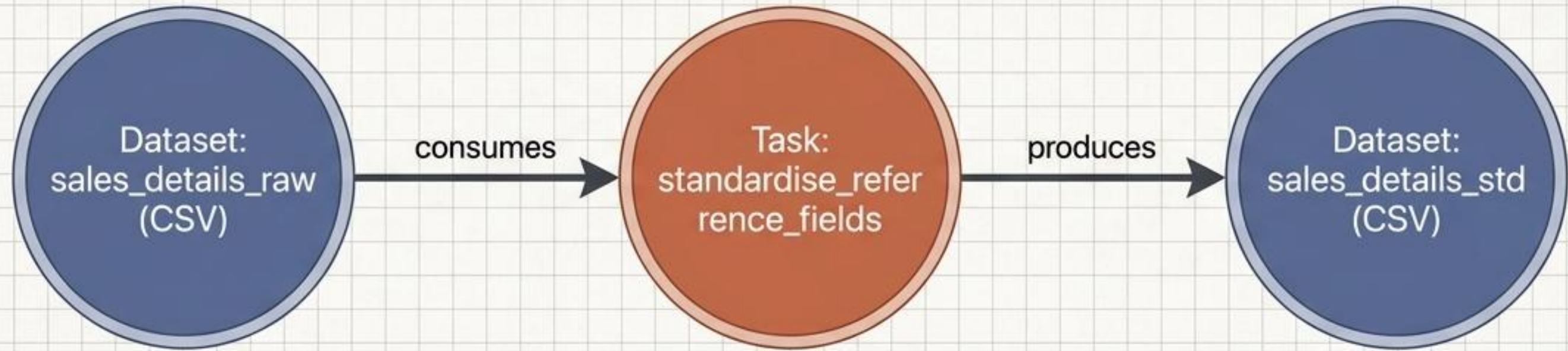
The dashboards, ML models, and stakeholders that rely on the data.

An enterprise-wide map linking state, action, and consumption to make data provenance explicit.

Anatomy Part I: Datasets & Platforms



Anatomy Part II: Tasks & Data Pipelines



Source Dataset

The raw data source before transformation.

The Task (The Verb)

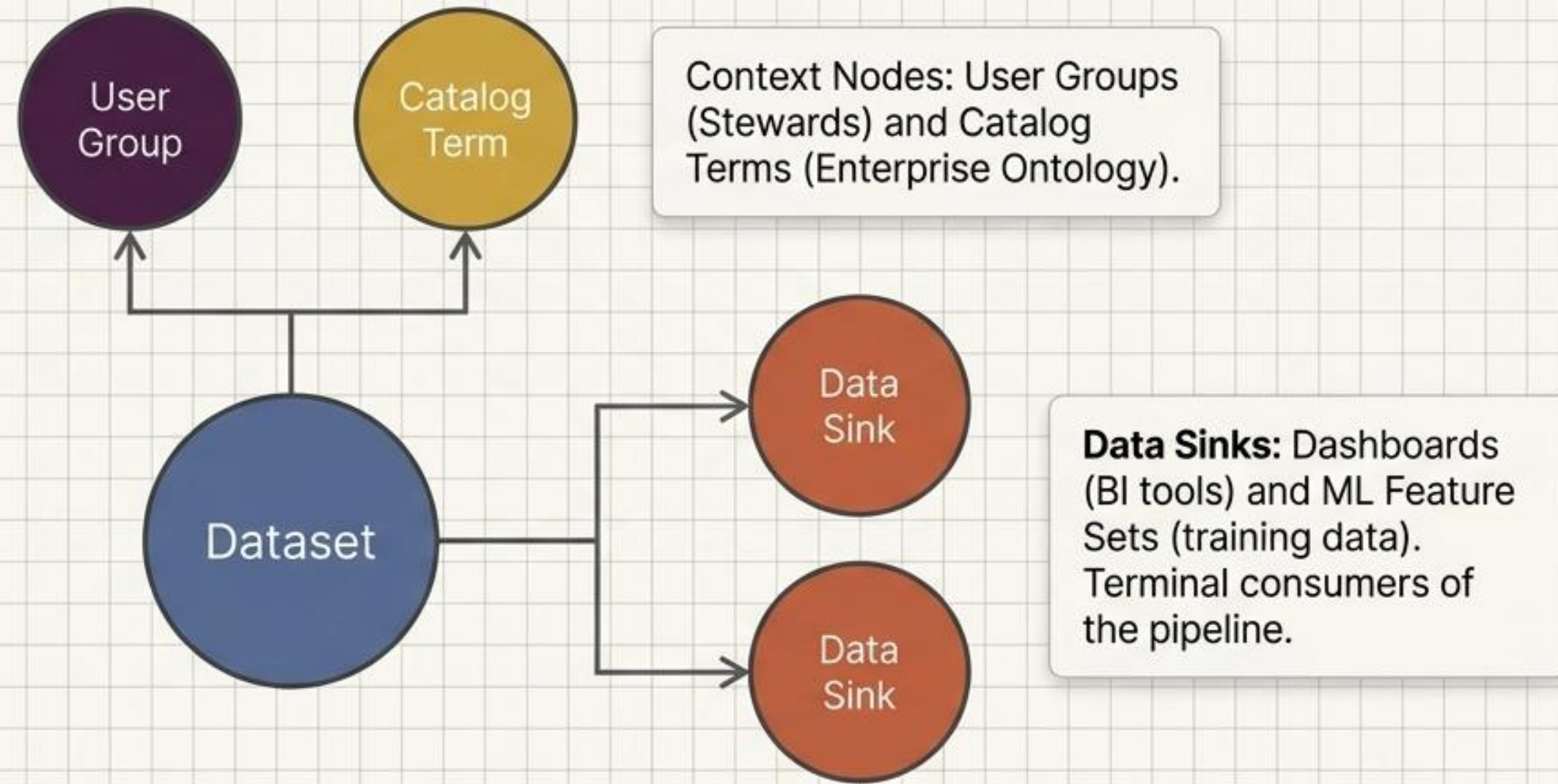
Explicitly maps dependencies by acting as the engine of data transformation.

Target Dataset

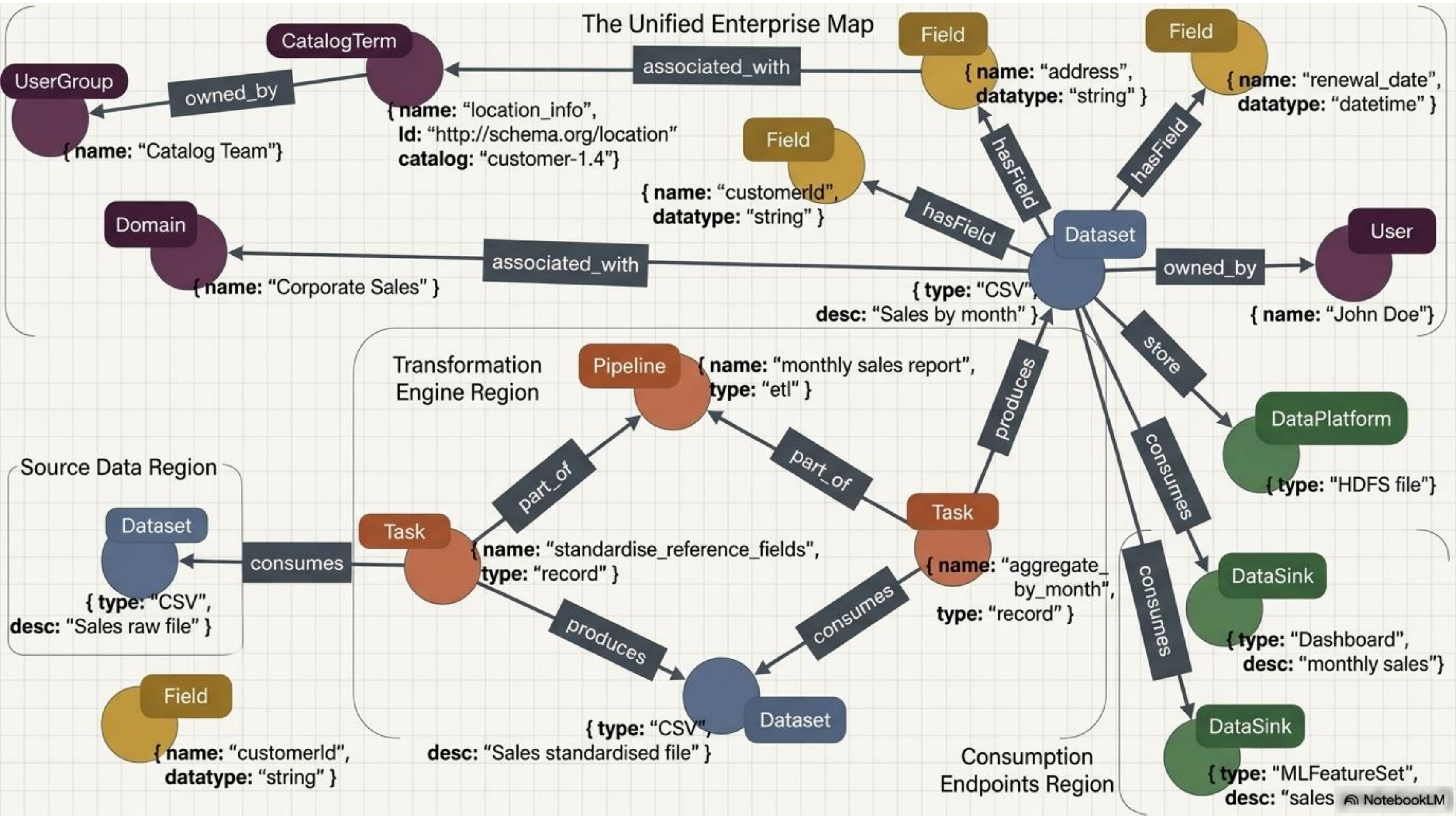
The resulting data after the task's operations.

Tasks act as the verbs of the enterprise map, making process dependencies explicit.

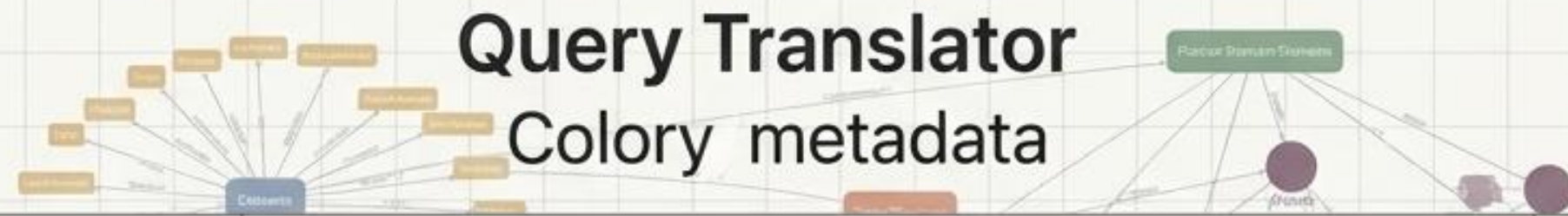
Anatomy Part II: Sinks, Stewards, and Domains



**Data without a sink is a cost;
data with a sink is an asset.**



Operationalizing the Graph: Cypher into Action



Business Question	Graph Logic	Value
What is the most popular corporate sales dataset?	(Dataset)<-[:consumes]-(DataSink)	Discovery
If this ETL task fails, who gets the broken report?	(Task)-[:produces]->(Dataset)<-[:consumes]-(DataSink)	Incident Response
Can I trust the data on this dashboard?	(DataSink)-[:consumes]->(Dataset)-[:source]->(Platform)	Auditing



Because data flows perfectly map to reality, graph traversal instantly answers complex business questions.

Use Case 1: Data Discovery (The Popularity Metric)

The Business Question

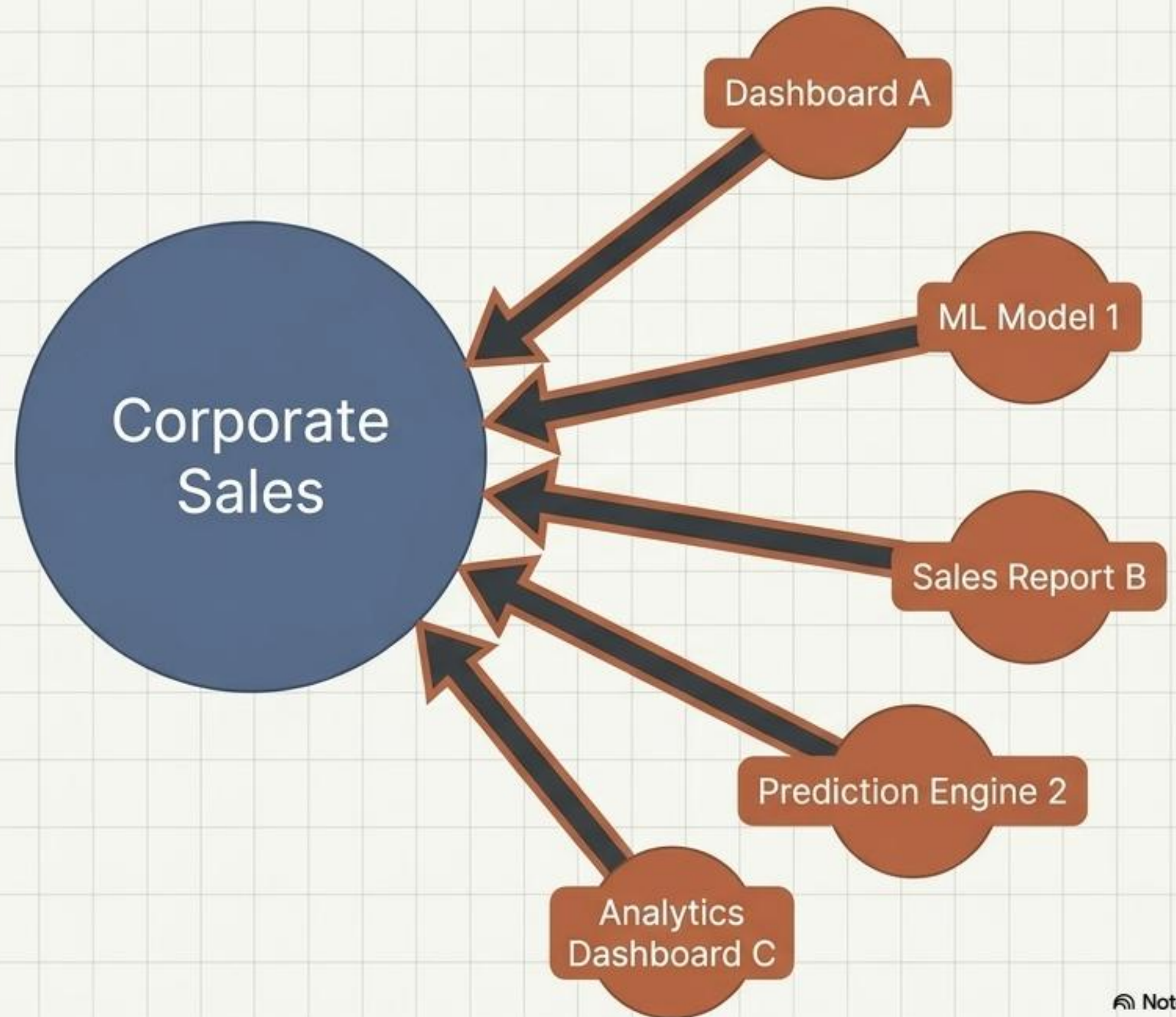
What is the most popular corporate sales dataset?

The Graph Mechanism

Count incoming [:consumes] relationships originating from terminal DataSinks.

The Operational Value

Empowers data scientists to bypass stale silos and instantly locate highly utilized, trusted corporate assets.



Use Case 2: Impact Analysis (The Domino Effect)

The Business Question

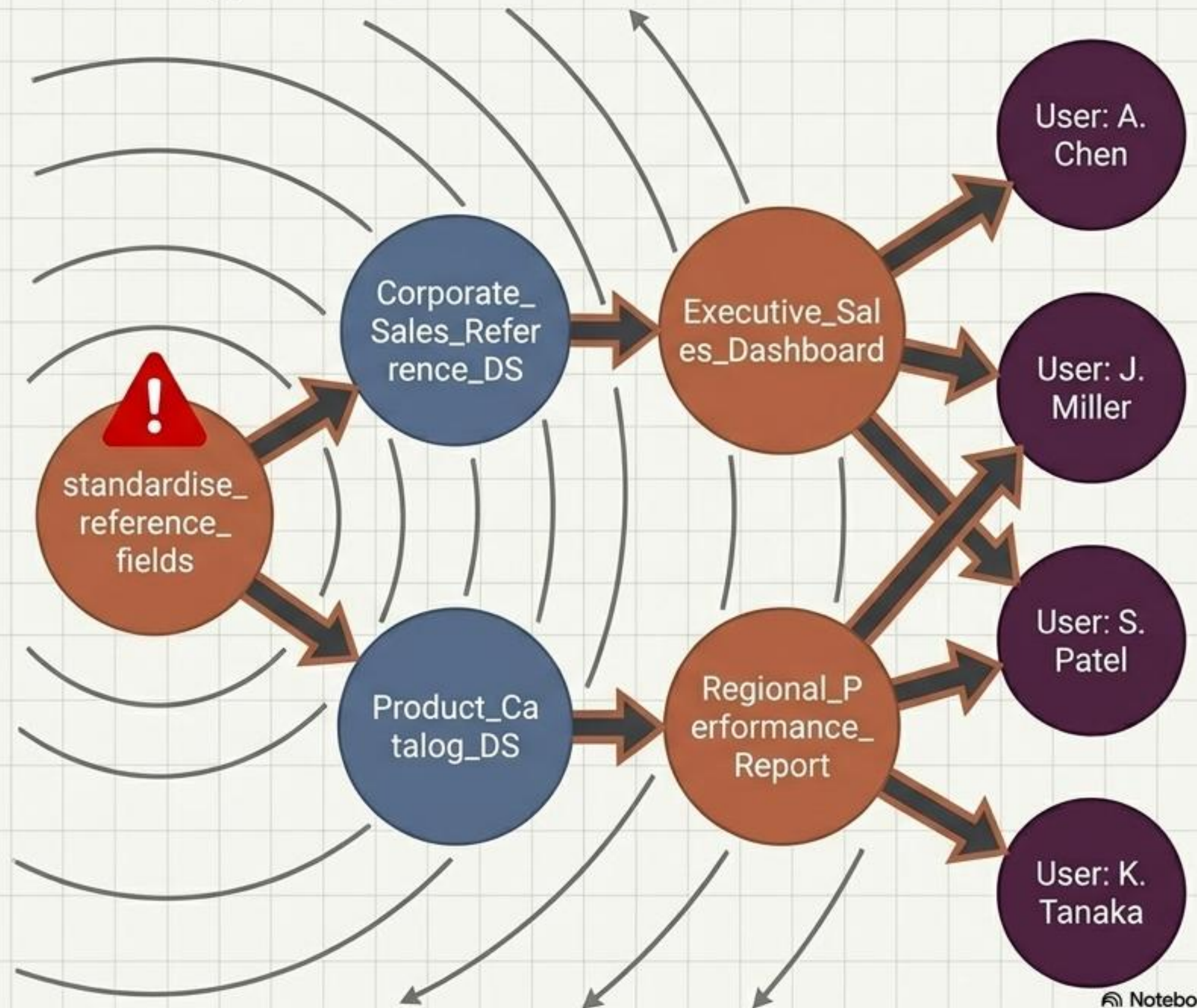
If this ETL task fails, who gets the broken report?

The Graph Mechanism

Traverse [:produces] and [:consumes] relationships strictly downstream to locate target sinks and their [:owned_by] relationships.

The Operational Value

Enables proactive alerting and rapid incident response before the business consumes bad data.



Use Case 3: Data Lineage (The Provenance Trace)

The Business Question

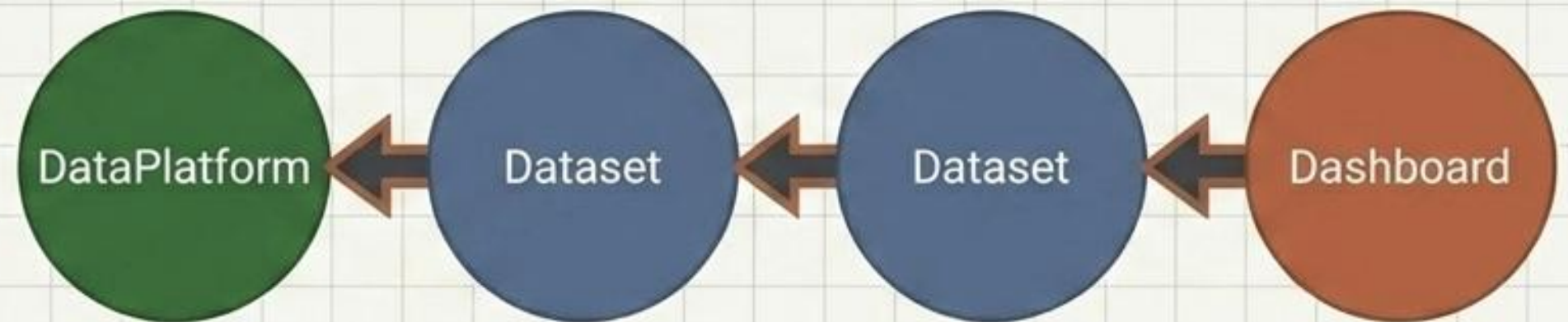
Can I legally and technically trust the data on this dashboard?

The Graph Mechanism

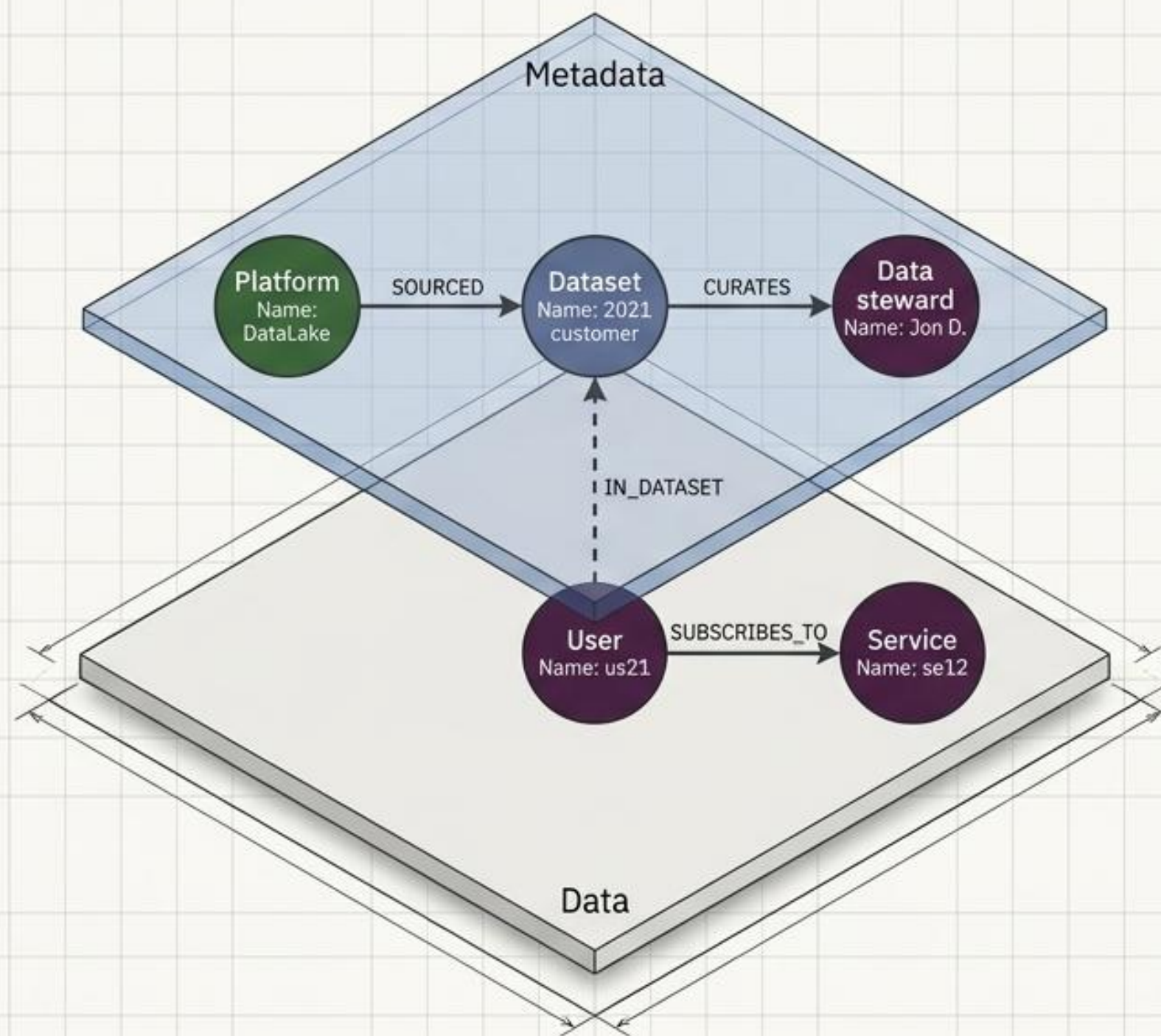
Traverse [:consumes] and [:produces] relationships strictly upstream to locate the ultimate [:source].

The Operational Value

Critical for regulatory compliance, debugging, and proving data integrity.



Synthesis Arc: The Dual-Layer Architecture



Non-Invasive Governance:
The metadata graph
hovers above raw
systems.

Outcome: Links disparate
partial views across the
enterprise without ever
disrupting underlying
source operations.

Diagnostic Matrix: Relational Catalogs vs. Graph Hubs

Relational Data Dictionaries	Metadata Knowledge Graphs
Flat Tables	 Highly Connected Networks
Manual/Siloed tracing	 Automated Multi-hop Traversal
Rigid Keyword matching	 Semantic and Relationship context
Passive Documentation	 Active Operational Intelligence

